3 Methodology

The aim of the thesis is to investigate the relation between linguistic typology and cross-lingual transfer. To this end, the methodology involves a typologically-informed, three-stage finetuning setup outlined in Section 3.2. The model – described in Section 3.1 – is finetuned for dependency parsing in a variety of languages, the selection of which is clarified in Section 3.3. Dependency parsing was chosen as a task for a number of reasons. Firstly, the Universal Dependencies (UD) [9] project provides a large number of uniform datasets in many different languages, facilitating cross-lingual comparisons on the same task. Secondly, to construct a correct dependency tree for a sentence in a given language, you need knowledge of the syntactic features of that language. Therefore, it is to be expected that dependency parsers necessarily encode language-specific syntactic information. How the results obtained from the setup serve to answer the research questions is explained in Section 3.4.

3.1 Model

The model used was implemented from the ground up, adapting Kondratyuk & Straka's [6] UDify model to better fit the setup of this thesis. In particular, an input token j is encoded using M-BERT [3] to obtain output $U_{i,j}$ at each layer $i \in [1 \ldots 12]$. A weighted sum r_j is then calculated as follows: Don't use lambda here

$$r_j = \eta \sum_{i=1}^{12} U_{i,j} \cdot \mathrm{softmax}(\lambda)_i$$

with η a trainable scalar and λ a vector of trainable scalar mixing weights. Next, r_j is passed through the classifier, classifiers [4] for predicting the dependency arc and its label. The input to the classifiers comes from four separate MLPs with 256 (arc) or 768 (dep) hidden dimensions and ELU non-linear activation:

$$\begin{aligned}
H_{\text{arc-head},j} &= \mathrm{ELU}(W_{\text{arc-head}} r_j + b_{\text{arc-head}}) \\
H_{\text{arc-dep},j} &= \mathrm{ELU}(W_{\text{arc-dep}} r_j + b_{\text{arc-dep}}) \\
H_{\text{tag-head},j} &= \mathrm{ELU}(W_{\text{tag-head}} r_j + b_{\text{tag-head}}) \\
H_{\text{tag-dep},j} &= \mathrm{ELU}(W_{\text{tag-dep}} r_j + b_{\text{tag-dep}})
\end{aligned}$$

The classifiers then score all possible dependency arcs:

$$\begin{aligned}
\mathcal{S}_{\text{arc}} &= H_{\text{arc-head},j} \mathbf{W}_{\text{arc}} H_{\text{arc-dep},j}^{\top} + \mathbf{b}_{\text{arc}} \\
\mathcal{S}_{\text{tag}} &= H_{\text{tag-head},j} \mathbf{W}_{\text{tag}} H_{\text{tag-dep},j}^{\top} + \mathbf{b}_{\text{tag}}
\end{aligned}$$

Finally, the Chu-Liu/Egmonds algorithm [2] is used to obtain the most probably dependency tree based on $\mathcal{S}_{\text{arc}}$ and $\mathcal{S}_{\text{tag}}$.

3.2 Finetuning Setup

As stated, the finetuning setup follows three stages where it first adapts to a source language, then optionally to one of four auxiliary languages, and finally to one of four target languages:

Stage 1: The goal of the first stage is to instill general knowledge of the task at hand. To this end, the model is finetuned on a treebank in the source language, German, resulting in the model M_{deu};
Stage 2: In the optional second stage, M_{deu} is finetuned on one of the auxiliary languages $\ell \in \{nld, nor, ces, fin\}$[1] to obtain four versions of M_ℓ;
Stage 3: M_{deu} and each M_ℓ is finetuned on each of the target languages $\lambda \in \{gsw, fao, hsb, vep\}$,[2] resulting in 4 instances of $M_{deu,\lambda}$ and 16 instances of $M_{\ell,\lambda}$.

In Stage 1, the model is finetuned over the course of 50 epochs on the full German training set with the Adam optimizer [5] and a cosine-based learning rate scheduler with 15 % warmup. In Stages 2 & 3, the training sets are matched in size by randomly sampling a fixed number of examples from each training set. In the second stage, the model is finetuned on 10 000 examples from each of the auxiliary languages' treebanks over the course of 100 epochs. In the third stage, the model is finetuned on 50 examples from each of the target langauges' treebanks over the course of 2 epochs. An overview of the hyperparameters used is given in Appendix A.This is not currently true

3.3 Language Selection

Broadly speaking, language selection is based on their representation in M-BERT's pre-training data as reported by Wu & Dredze [11] and their typological similarity. Here, typological similarity is quantified as the cosine similarity between each language's binary `syntax_knn` feature vector in the URIEL knowledge base [8]. Each dimension of that vector indicates the presence or absence of a syntactic property. An overview of the selected languages and their respective treebanks is shown in Table 3.1.

German is selected as the source language because it is one of the most well-represented languages in M-BERT's pre-training data. Moreover, Turc et al. [10] set out to find the best source language for cross-lingual transfer by finetuning models on various monolingual datasets for a wide variety of tasks and measuring their zero-shot performance on different-language test sets. They found that models trained on German (as well as Russian) performed best, further motivating the selection of German as the source language in this thesis.

The auxiliary languages are all moderately and roughly equally well-represented in M-BERT's training data. Their selection is motivated phylogenetically, as they each represent a different level of relatedness to German:

- Finnish is part of the Uralic language family and is entirely unrelated to German;

[1] Dutch, Norwegian, Czech, and Finnish.
[2] Swiss German, Faroese, Upper Sorbian, and Veps.

Language	Family	ISO 639-3	Treebank	No. of tokens	No. of sentences
Czech	West Slavic	ces	CAC	494 142	24 709
Dutch	West Germanic	nld	Alpino	208 745	13 603
Faroese	North Germanic	fao	OFT	10 002	1 208
Finnish	Finnic	fin	TDT	202 210	15 136
German	West Germanic	deu	HDT	3 399 390	189 928
Norwegian	North Germanic	nor	Bokmaal	310 221	20 044
Swiss German	West Germanic	gsw	ATB	652	98
Upper Sorbian	West Slavic	hsb	UFAL	11 196	646
Veps	Finnic	vep	VWT	1 303	103

Table 3.1: The languages and their specific UD treebanks used in this thesis.

- Czech, like German, is part of the Indo-European language family. However, their relatedness is limited as they are each part of a different branch of the language family: Czech is part of the Slavic branch, whereas German is part of the Germanic branch;
- Norwegian is also part of the Germanic branch of the Indo-European language family, but it is part of the North Germanic subbranch, whereas German is part of the West Germanic subbranch;
- Dutch is very closely related to German as both languages are West Germanic languages.

A hierarchical clustering analysis of the `syntax_knn` vectors found these levels of relatedness to be reflected in the languages' mutual typological similarities, too: Font should be changed/enlarged, ISO code for Norwegian is w

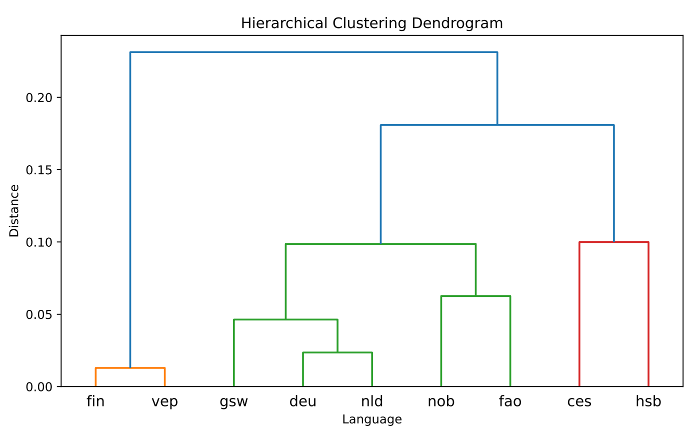

Fig. 3.1: Agglomerative Hierarchical Clustering dendrogram based on the cosine similarity of the `syntax_knn` vectors of the included languages.

The target languages are selected by finding the most typologically similar language to each of the auxiliary languages that have a UD treebank but are not represented in M-BERT's pre-training data. For Dutch, the most similar language to meet the criteria is Swiss German, Norwegian is matched with Faroese, Czech with Upper Sorbian, and Finnish with Veps. The treebanks of the low-resource languages are randomly split into separate train, test, and validation sets as their treebanks do not contain these splits themselves.

3.4 Experiments

The research questions can be classified as evaluating the success of the setup or, more broadly, of typology-guided cross-lingual transfer (**Q1**, **Q2**, **Q3**) and as investigating how shifts in the model parameters facilitate cross-lingual transfer (**Q4**). Section 3.4.1 outlines how the former group is answered, Section 3.4.2 does the same for the latter.

3.4.1 Model Evaluation

Successful transfer is defined as higher performance on the target language test set. Following the example of Lauscher et al. [7], the unlabeled attachment score (UAS) is used to quantify performance.[3] In this section, the following notations are used:

l	Any arbitrary language.
ℓ	Any arbitrary auxiliary language (Dutch, Norwegian, Czech, or Finnish).
λ	Any arbitrary target language (Swiss German, Faroese, Upper Sorbian, or Veps).
M_l	A model finetuned on language l. If $l = \ell$, it is implied that it was first finetuned on German, as there are no cases where a model is directly finetuned on ℓ in this setup.
M_{l_1,l_2}	A language trained on l_1, then on l_2.
$\mathrm{s}(l_1, l_2)$	Cosine similarity between the `syntax_knn` vectors of l_1 and l_2.
$\mathrm{UAS}(M_{l_1}, l_2)$	The performance (in UAS) of M_{l_1} when evaluated on the test set of l_2.
$\Delta_{l_3}(M_{l_1}, M_{l_2})$	$\mathrm{UAS}(M_{l_1}, l_3) - \mathrm{UAS}(M_{l_2}, l_3)$, i.e., the performance of M_{l_2} on l_3 subtracted from the performance of M_{l_1} on l_3. The value is positive when M_{l_1} performs better on l_3, negative when M_{l_2} performs better on l_3.

Q1: Does greater typological similarity between the source and target languages result in better zero-shot cross-lingual transfer performance?
If typological similarity indeed leads to better zero-shot performance, the expectation is that $\Delta_\lambda(M_\ell, M_{deu})$ is positive when $\mathrm{s}(\ell, \lambda) > \mathrm{s}(\mathrm{deu}, \lambda)$. That is, a model finetuned only on German has worse zero-shot performance on λ than a model that is finetuned on an ℓ that is typologically similar to λ. For example, $\Delta_{fao}(M_{nor}, M_{deu})$ is expected to be positive because Norwegian is more typologically similar to Faroese than German is.

[3] Using the more widely reported LAS would inhibit comparisons between datasets, as different treebanks may have been labeled with different annotation schemes.

Q2: To what degree does training on a few examples improve performance over zero-shot performance?
While Lauscher et al. [7] found training on as few as 10 examples to yield considerable improvements over zero-shot performance already, it is important to quantify the degree to which that holds in this setup. After Stage 3, the models M_{deu} and M_ℓ are finetuned on 100 examples from λ. The result is twenty models $M_{l,\lambda}$. For each of those, improvement over zero-shot performance is defined as $\Delta_\lambda(M_{l,\lambda}, M_l)$. Since training on a language is not expected to be detrimental to performance on that language, this value is not expected to ever be negative.

Q3: Is training on a few examples from a target language more effective when first finetuning on a typologically similar auxiliary language?
This question can effectively be split into two components. Both are answered by investigating whether $\Delta_\lambda(M_{l_1,\lambda}, M_{l_2,\lambda})$ is positive whenever $\mathrm{s}(l_1, \lambda) > \mathrm{s}(l_2, \lambda)$. That is, performance on λ is higher when the model finetuned to a language similar to λ in the preceding stage. One component of the question pertains to the effectiveness of the auxiliary stage in general. In this case, l_1 is any auxiliary language ℓ and l_2 is German. When ℓ is more similar to λ than German is, the second stage may provide the model with new typological information that can then be leveraged in the third stage. The other component of the question pertains to the typological similarity between the auxiliary language and the target language. In this case, l_1 is any given auxiliary langauge and l_2 is any of the other auxiliary languages. This serves to assess whether any improvements found are the result of typology, rather than longer training in general.

3.4.2 Probing Experiments

Following Choenni & Shutova [1], investigation of the models' parameters is done using probing. That is, a classifier is trained on the embeddings created by the model to investigate what these embeddings encode. Choenni & Shutova [1] do this by training a one-hidden-layer MLP on various sentence embeddings to detect whether certain typological features are encoded in the embedding.

In this thesis, a similar approach is used, but instead training a probe to detect which language an input sentence was written in given a sentence embedding. To this end, a dataset is compiled using 404 random sentences from each of the nine languages under scrutiny in this thesis. These sentences taken from the Tatoeba corpora (available at `https://tatoeba.org`). One half of each language's sentences is used in the training set, the other half is used in the test set. For each sentence, its tokens are passed through the M-BERT encoder of a given dependency parser. After each layer, mean-pooling is applied over all token embeddings to obtain a sentence embedding. This sentence embedding is then passed through a 100-dimensional layer, followed by a softmax output layer to predict which language the input sentence was written in.

Q4: Does finetuning on a language make its typological features more easily identifiable in the model's embedding space?

For this question, identifiability of language l_1 is defined by how accurately a probe can detect whether a given sentence was written in language l_1 from the sentence's embedding. If finetuning on language l_1 makes its typological features more easily identifiable in the model's embedding space, it stands to reason that the accuracy of the probe in detecting l_1 consequently goes up. Moreover, a typologically similar l_2 has many of the same typological features. Therefore, if l_1 becomes more easily identifiable because its typological features are more clearly encoded, probe accuracy on detecting l_2 should go up as well.

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A Appendix